



Note. A special counter named ICOUNT is available from a manufacturer in Madurai. This counter requires only one port pin from Arduino. Reset and latch signals are automatically generated by ICOUNT. The number to be displayed is transferred from any available output port of Arduino as a train of pulses. When pulses stop to arrive, the display is updated.

Circuit diagram of the line-frequency meter with reciprocal counting is shown in Fig. 7. A 230V primary to 12V-0-12V, 300mA secondary step-down transformer (X1) is used for the power supply input and to get a signal to drive transistor BC547 (T1) that provides a square-wave output corresponding to line-frequency input.

The code (LFM.ino) is written in Arduino programming language. Tone() function produces a 5kHz clock pulse at pin 8 of Board1, and this is connected to Timer1 input at pin 5. Clock pulse train from pin

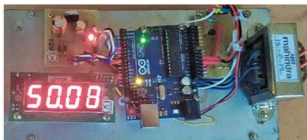
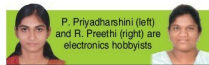


Fig. 8: Author's prototype

12 of Board1 is given to the four-digit counter display module. The +5V supply to the display module is taken from Arduino (Board1) itself as the current needed is very small. The author's prototype with 50.08Hz line frequency having a precision of 0.01Hz is shown in Fig. 8.

ATmega328 is programmed using Arduino IDE software. Select the correct board from Tools→Board menu in Arduino IDE, and burn the program (sketch) through the standard USB port in your computer. **EFY**



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